BESS HAGAN

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SUMMARY

Data analyst with a 4.0 GPA Computer Science degree, hands-on experience building a SQL-to-Python data pipeline, and a peer-reviewed publication at the 21st AAAI Conference on AI and Interactive Digital Entertainment (AIIDE-25). Skilled at cleaning and transforming complex datasets, applying EDA and statistical methods, and translating findings into actionable insights. Supported NSF-funded research through rigorous data wrangling and behavioral analysis across 100+ user records.

TECHNICAL SKILLS

Data Analysis & Visualization: Python (Pandas, NumPy, Matplotlib), SQL, R, Excel (Pivot Tables, Dashboards), Exploratory Data Analysis (EDA), Data Cleaning & Transformation, Hypothesis Testing, Statistical Analysis

Databases & Tools: PostgreSQL, pgAdmin, Data Pipelines, Jupyter Notebook, Git/GitHub, VS Code, Anaconda

Machine Learning: scikit-learn, PyTorch, Linear & Logistic Regression, Decision Trees, KNN, K-Means

WORK EXPERIENCE

Student Researcher | SURF 2025 (Procedural Content Generation) *May 2025 – Jul 2025*

Southwestern University, Georgetown, TX

- **Publication:** Coauthored a peer-reviewed paper accepted and presented at the **21st AAAI Conference on AI and Interactive Digital Entertainment (AIIDE-25)**.
- Trained conditional and unconditional diffusion models with UNet architecture in PyTorch for procedural game level generation; implemented cross-entropy loss and early stopping within an existing codebase.
- Developed scripts to evaluate and visualize A* solvability metrics across models, and created batch scripts for data preparation, training, and results reporting.

Student Researcher | NSF-funded research (DRL-2229612) *May 2024 – Aug 2024*

INVITE Institute/University of Illinois Urbana-Champaign, Champaign, IL

- Analyzed behavioral log data from ~100 junior high students across 62 user accounts in a Minecraft-based learning environment to identify markers of student persistence and struggle.
- Designed and built a SQL-to-Python data pipeline to extract, clean, and transform event log data from a multi-table relational database, including coordinate logs, timestamped button presses, and code block interaction records, into structured binary event logs for analysis.
- Designed operational definitions for 10+ in-game event types (e.g., puzzle completion state, run and reset event counts, Levenshtein edit distance between solution attempts) to transform raw log data into a structured, analysis-ready format.
- Investigated and resolved data quality discrepancies between the application and database layers in collaboration with developers.

PROJECTS

E-Commerce Sales Analysis | Python, Pandas, Matplotlib, Excel Pivot Tables *Portfolio Write-Up*

- Analyzed ~14,000 e-commerce transactions to identify revenue drivers and purchasing behavior patterns, addressing five defined business questions.
- Performed full analysis in Excel (Pivot Tables, charts) then reproduced in Python with Pandas and Matplotlib to validate results and assess workflow trade-offs.
- Found that revenue is driven primarily by repeat purchase frequency rather than discounting, device type, or delivery speed; recommended retention-focused growth strategies over short-term tactical optimizations.

EDUCATION

B.A. Computer Science | Minor in Data Analytics *Dec 2025*

Southwestern University, Georgetown, TX — GPA: 4.0, Summa Cum Laude

- Chapter President, Upsilon Pi Epsilon National Honorary Computer Science Society (2024–2025)

A.S. Computer Science *May 2023*

Temple College, Temple, TX — GPA: 4.0

- Vice President, Lambda Theta Chapter, Phi Theta Kappa Honor Society (2022–2023)